

Unit 2

ER-to-Relational mapping

Steps :

1) Mapping of Regular Entity Types

Employee

Frame	mno	Lname	<u>Ssn</u>	Bdate	address	Sex	Salary
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department

Dname	<u>Dnumber</u>
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Project

Pname	<u>Pno.</u>	Plocation
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Step 2 :

Mapping of Weak Entity Type

dependent.

<u>Essn</u>	d-name	Sex	Bdate	Relationship
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FK

* Include - foreign key attribute of R, PK - that corresponds to owner entity.

FK+d-name → PK for dependent table.

Steps

Mapping of Binary 1:1 Relationship Types

- For each binary 1:1 relationship R , in ER schema, identify relations S and T that correspond to entity types participating in R .
- Choose one of relation say S - include PK in S the PK of T . It is better to choose an entity type with total participatn. in R .
- In COMPANY ER diagram $\rightarrow$ manages - have 1:1 relationship, so we need to convert manages 1-to-1 relationship.
- In manages - two entities are participating.
- We need to modify any one of two table.
 - Choose with total participatn, i.e. DEPARTMENT.
 - So include PK of emp as FK to department.
- Hence.

DEPARTMENT

Dname	<u>Dno.</u>	Mgr-ssn	
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- Also we have one attribute in relationship manages we add start-date to the same table.

So, DEPARTMENT

Dname	<u>Dno.</u>	Mgr-ssn	Mgr-start-date
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Step 4: Mapping of Binary 1:N Relationship

Types

In COMPANY ER diagram, we have 3 1-to-many relationship →

- 1) Supervision
- 2) Controls
- 3) works-for

we need to modify 'N' side table.

1) for works-for → Employee table should be modified

2) Controls → project table should be modified.

3) Supervision →

① We need to modify employee table, make PK of dept as FK in employee table

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Sal	<u>Dno</u>	Supassn
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work-for - ✓ Supervision - ✓

② We need to modify project table - So include, PK of dept as FK in project table.

PROJECT

Pname	<u>Pno</u>	Plocation	Dno
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- For these 3 relations, there is no attributes in relations $\rightarrow$ hence there is no inclusion of attributes.

Step 5:

Mapping of Binary m:n relationship types.

- We have one many-to-many relationship

WORKS-ON.

- For this we need to create a table, with two PK - SSN and PNO, and add relationship attribute hours in newly created table.

So,

WORKS-ON

<u>SSN</u>	<u>PNO</u>	Hours
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Step 6:

Mapping of multivalued attribute

- department location is multivalued attribute, so we need to create new table and we need to include PK of this dept in newly created table.

DEPT-LOCATIONS

<u>PNO</u>	<u>Dlocation</u>
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Combination of these two keys is PK.

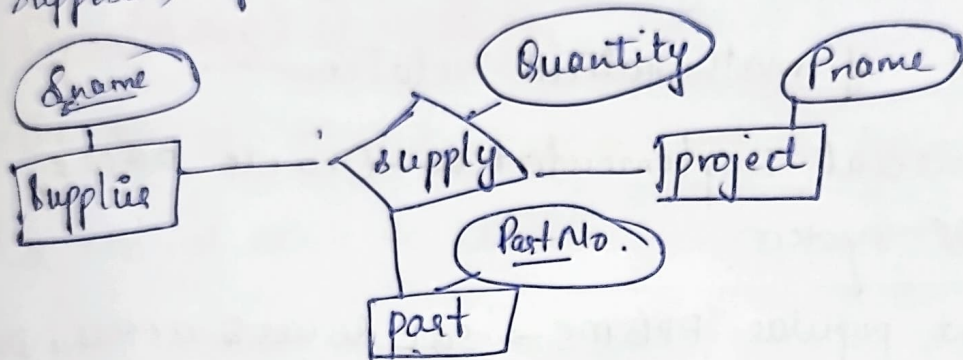
step 7:

mapping of N-ary Relationship Types.

(for n-ary)

- We don't have any Ternary relationship in COMPANY ER diagram.

Suppose, if we have



- For supply-relationship, we need to create all the table and we need to include PK of all participating entities.

SUPPLIER

<u>Sname</u>	...
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PROJECT

<u>Pname</u>	
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PART

<u>Pno</u>	...
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SUPPLY

<u>Sname</u>	<u>Pname</u>	<u>Pno</u>	Quantity
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- Combination of Sname + Pname + Pno will be PK for this newly created table Supply.